Category
Introduction to Big Data Techniques

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HOW IS FINTECH USED IN QUANTITATIVE INVESTMENT ANALYSIS?

Learning Outcomes

- describe aspects of "fintech" that are directly relevant for the gathering and analyzing of financial data.
- describe Big Data, artificial intelligence, and machine learning

In its broadest sense, the term **fintech** generally refers to technology-driven innovation occurring in the financial services industry. For our purposes, fintech refers to technological innovation in the design and delivery of financial services and products. In common usage, fintech can also refer to companies involved in developing the new technologies and their applications, as well as the business sector that includes such companies. Many of these innovations are challenging the traditional business models of incumbent financial services providers.

Early forms of fintech included data processing and the automation of routine tasks. Systems that provided execution of decisions according to specified rules and instructions followed. Fintech has advanced into decision-making applications based on complex machine-learning logic, in which computer programs are able to "learn" how to complete tasks over time. In some applications, advanced computer systems are performing tasks at levels that far surpass human capabilities. Fintech has changed the financial services industry in many ways, giving rise to new systems for investment advice, financial planning, business lending, and payments.

Whereas fintech covers a broad range of services and applications, areas of development that are more directly relevant to quantitative analysis in the investment industry include the following:

- **Analysis of large datasets.** In addition to growing amounts of traditional data, such as security prices, corporate financial statements, and economic indicators, massive amounts of **alternative data** generated from non-traditional data sources, such as social media and sensor networks, can now be integrated into a portfolio manager's investment decision-making process and used to help generate alpha and reduce losses.
- **Analytical tools.** For extremely large datasets, techniques involving **artificial intelligence (AI)**—computer systems capable of performing tasks that previously required human intelligence—might be better suited to identify complex, non-linear relationships than traditional quantitative methods and statistical analysis. Advances in AI-based techniques are enabling different data analysis approaches. For example, analysts are turning to AI to sort through the enormous amounts of data from company filings, annual reports, and earnings calls to determine which data are most important and to help uncover trends and generate insights relating to human sentiment and behavior.

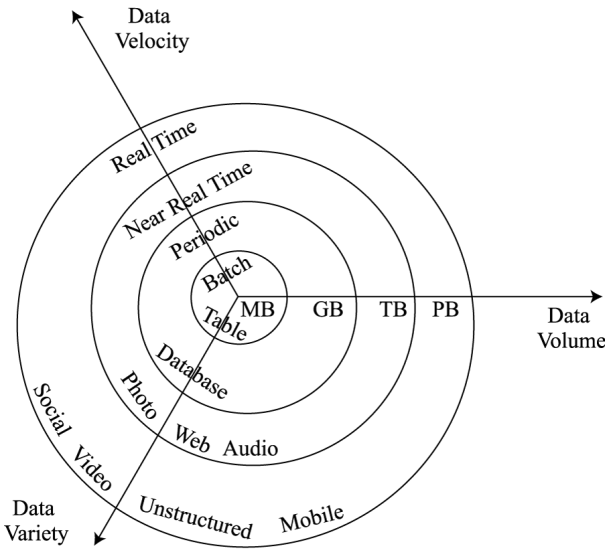
Big Data

As noted, datasets are growing rapidly in terms of the size and diversity of data types that are available for analysis. The term **Big Data** has been in use since the late 1990s and refers to the vast amount of information being generated by industry, governments, individuals, and electronic devices. Big Data includes data generated from traditional sources—such as stock exchanges, companies, and governments—as well as non-traditional data types, also known as alternative data, arising from the use of electronic devices, social media, sensor networks, and company exhaust (information generated in the normal course of doing business).

Traditional data sources include corporate data in the form of annual reports, regulatory filings, sales and earnings figures, and conference calls with analysts. Traditional data also include data that are generated in the financial markets, including trade prices and volumes. Because the world has become increasingly connected, we can now obtain data from a wide range of devices, including smart phones, cameras, microphones, radio-frequency identification (RFID) readers, wireless sensors, and satellites that are now in use all over the world. As the internet and the presence of such networked devices have grown, the use of non-traditional data sources, or alternative data sources—including social media (posts, tweets, and blogs), email and text communications, web traffic, online news sites, and other electronic information sources—has risen.

The term *Big Data* typically refers to datasets that have the following characteristics:

- **Volume**: The amount of data collected in files, records, and tables is very large, representing many millions, or even billions, of data points.
- **Velocity**: The speed and frequency with which the data are recorded and transmitted has accelerated. Real-time or near-real-time data have become the norm in many areas.
- **Variety**: The data are collected from many different sources and in a variety of formats, including structured data (e.g., SQL tables), semistructured data (e.g., HTML code), and unstructured data (e.g., video messages).



Data	Volume Key	Bytes of Information
MB	Megabyte	One Million
GB	Gigabyte	One Billion
TB	Terabyte	One Trillion
PB	Petabyte	One Quadrillion

Source: Ivy Wigmore, "Definition: 3Vs (Volume, Variety and Velocity)," WhatIs.com, last updated December 2020, http://whatis.techtarget.com/definition/3Vs.

Exhibit 1 shows that data volumes are growing from megabytes and gigabytes to far larger sizes, such as terabytes and petabytes, as more data are being generated, captured, and stored. At the same time, more data, traditional and non-traditional, are available on a real-time or near-real-time basis with far greater variety in data types than ever before.

When Big Data is used for inference or prediction, a "fourth V" comes into play— veracity—which relates to the credibility and reliability of different data sources. Determining the credibility and reliability of data sources is an important part of any empirical investigation. The issue of veracity becomes critically important for Big Data, however, because of the varied sources of these large datasets. Big Data amplifies the age-old challenge of disentangling quality from quantity.

Big Data can be structured, semi-structured, or unstructured data. Structured data items can be organized in tables and are commonly stored in a database where each field represents the same type of information. Unstructured data can be disparate, unorganized data that cannot be represented in tabular form. Unstructured data, such as those generated by social media, email, text messages, voice recordings, pictures, blogs, scanners, and sensors, often require different, specialized applications or custom programs before they can be useful to investment professionals. For example, to analyze data contained in emails or texts, specially developed or customized computer code might be required to first process these files. Semistructured data can have attributes of both structured and unstructured data.

Sources of Big Data

Big Data, therefore, encompasses data generated by the following:

- financial markets (e.g., equity, fixed income, futures, options, and other derivatives),
- businesses (e.g., corporate financials, commercial transactions, and credit card purchases),
- governments (e.g., trade, economic, employment, and payroll data),
- individuals (e.g., credit card purchases, product reviews, internet search logs, and social media posts),
- sensors (e.g., satellite imagery, shipping cargo information, and traffic patterns), and, in particular,
- the Internet of Things, or IoT (e.g., data generated by "smart" buildings, where the building is providing a steady stream of information about climate control, energy consumption, security, and other operational details).

In gathering business intelligence, historically, analysts have tended to draw on traditional data sources, using statistical methods to measure performance, predict future growth, and analyze sector and market trends. In contrast, the analysis of Big Data incorporates the use of alternative data sources.

From retail sales data to social media sentiment to satellite imagery that might reveal information about agriculture, shipping, and oil rigs, alternative datasets can provide additional insights about consumer behavior, firm performance, trends, and other factors important for investment-related activities. Such

- data generated by individuals,
- data generated by business processes, and
- data generated by sensors.

Data generated by individuals are often produced in text, video, photo, and audio formats and also can be generated through such means as website clicks or time spent on a webpage. This type of data tends to be unstructured. The volume of this type of data is growing dramatically as people participate in greater numbers and more frequently in online activities, such as social media and e-commerce, including online reviews of products, services, and entire companies, and as they make personal data available through web searches, email, and other electronic trails.

Business process data include information flows from corporations and other public entities. These data tend to be structured data and include direct sales information, such as credit card data, as well as corporate exhaust. Corporate exhaust includes corporate supply chain information, banking records, and retail point-of-sale scanner data. Business process data can be leading or real-time indicators of business performance, whereas traditional corporate metrics might be reported only on a quarterly or even yearly basis and typically are lagging indicators of performance.

Sensor data are collected from such devices as smart phones, cameras, RFID chips, and satellites that usually are connected to computers through wireless networks. Sensor data can be unstructured, and the volume of data is many orders of magnitude greater than that of individual or business process datastreams. This form of data is growing exponentially because microprocessors and networking technology are increasingly present in a wide array of personal and commercial electronic devices. Extended to office buildings, homes, vehicles, and many other physical forms, this culminates in a network arrangement, known as the **Internet of Things**, which is formed by the vast array of physical devices, home appliances, smart buildings, vehicles, and other items that are embedded with electronics, sensors, software, and network connections that enable the objects in the system to interact and share information.

Exhibit 2 shows a classification of alternative data sources and includes examples for each.

Exhibit 2: Classification of Alternative Data Sources

Individuals	Business Processes	Sensors
Social media	Transaction data	Satellites
News, reviews	Corporate data	Geolocation
Web searches, personal data		Internet of Things
		Other sensors

In the search to identify new factors that could affect security prices, enhance asset selection, improve trade execution, and uncover trends, alternative data are being used to support data-driven investment models and decisions. As interest in alternative data has risen, the number of specialized firms that collect, aggregate, and sell alternative datasets has grown.

Although the market for alternative data is expanding, investment professionals should understand the potential legal and ethical issues related to information that is not in the public domain. For example, the **scraping** of web data potentially could capture personal information that is protected by regulations or that might have been published or provided without the explicit knowledge and consent of the individuals involved. Best practices are still in development in many jurisdictions, and because of varying approaches taken by national regulators, the different forms of guidance could conflict.

Big Data Challenges

Big Data poses several challenges when it is used in investment analysis, including the quality, volume, and appropriateness of the data. Key issues revolve around the following questions, among others: Does the dataset have selection bias, missing data, or data outliers? Is the volume of collected data sufficient? Is the dataset well suited for the type of analysis? In most instances, the data must be sourced, cleansed, and organized before analysis can occur. This process can be extremely difficult with alternative data because of the unstructured characteristics of the data involved, which more often are qualitative (e.g., texts, photos, and videos) than quantitative in nature.

Given the size and complexity of alternative datasets, traditional analytical methods cannot always be used to interpret and evaluate these datasets. To address this challenge, AI and machine learning techniques have emerged that support work on such large and complex sources of information.

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